

Privacy-Preserving Collaborative Intrusion Detection Systems: A Federated Learning Framework

Chukwuebuka Ezelu and Ulrich Buehler

Department of Applied Computer Science, Fulda University of Applied Sciences, Germany

Email: {chukwuebuka.ezulu, u.buehler}@cs.hs-fulda.de

Abstract—Security concerns are one of the significant obstacles to collaborative systems. With the ever-increasing sophistication of attacks in networked environments, it is impossible to overlook the need for collaboration. Collaborative Intrusion Detection Systems (CIDS) presents a solution to combating such attacks but also introduces some privacy concerns. The need for information transmission within CIDS architecture raises some privacy issues, which can be deterring to organizations with sensitive data. This work presents a forest-based Federated Learning framework that allows organizations to collaboratively build a forest without compromising the security and privacy of their data. Two strategies were proposed for building the forest and were evaluated against a centralized dataset. The result shows that the proposed framework outperforms the centralized dataset framework based on model reliability and accuracy of prediction.

Index Terms—Collaborative Intrusion Detection Systems, Federated Learning, Data Privacy, Tree-based Classifiers

I. INTRODUCTION

Collaboration between IT infrastructures has become a cornerstone upon which modern research on network security and privacy depends. In many collaborative environments, such as Vehicular Ad Hoc Networks (VANET) [1] where Autonomous Driving Vehicles (ADV) communicates with Roadside Units (RSU) to provide traffic and other information, and Wireless Sensor Networks (WSN) [2] where Sensor Node (SN), Cluster Head (CH), and Base Station (BS) communicate with each other, sensitive information is transmitted within the architecture. Additionally, information is transmitted from one unit to the other in a collaborative network infrastructure. In recent years, Collaborative Intrusion Detection Systems (CIDS) has emerged as an effective tool for securing such a networked environment. However, just like the others, CIDS is plagued with the need to transmit information to different sections of the architecture. A centralized or traditional machine learning framework requires data from all the participating members of the collaboration to be collated on a central server for training. However, this creates a security concern regarding the privacy and sensitivity of the data to be transmitted. A member of the collaboration can be hesitant about transferring sensitive information to a third party, regardless of how secure it is. Federated Learning (FL) provides a framework for training a collaborative machine learning model without the need for exposing sensitive information [3]. There are two major categories of FL, namely Horizontal Federated Learning (HFL) and Vertical Federated Learning (VFL). HFL entails a scenario

where participating members' datasets are homogeneous (i.e., they share the same feature set but have different experiences or observations). VFL has heterogeneous members (i.e., they have similar observations but complementary features).

In this paper, the proposed system is a CIDS that adopts an HFL framework to preserve the privacy of sensitive information in the architecture. Furthermore, FL learns from a shared model, thereby eliminating the need for transmitting network data in the architecture. Further discussion on the system can be found in Section III-C. The rest of the paper is structured as follows. First, section II presents some works that adopt Federated Learning to improve data privacy. Next, section III describes the proposed framework in detail. Then, section IV evaluates the performance of the framework. Finally, section V concludes the work.

II. RELATED WORK

Liu et al. present multi-layer perceptron neural network-based federated learning for training Intrusion Detection Systems (IDS) in Autonomous Vehicular Network (AVN) in [4]. The FL strategy adopted by the authors is model update averaging. The ADVs transmit their locally trained model to the RSU, which computes the average of all the models, resulting in an aggregated model. However, neural networks are less effective on tabular and unbounded datasets than a tree-based model. Nonetheless, adopting model averaging in tree-based models is not feasible.

SecureBoost, proposed by Cheng et al. in [5], shows a lossless gradient-tree boosting federated learning framework that adopts Vertical Federated Learning strategy. The authors constructed a boosting tree over multiple participating entities with the same observations. This obtained the same accuracy as a non-federated, trained model with a centralized dataset. However, it requires an additional global model to be partially trained by each participating entity or member of the collaborative system.

Tian et al. propose FederBoost in [6], which is two distinct systems for VFL and HFL. Vertical FederBoost, as the name implies, handles vertically partitioned data, whereas horizontal FederBoost handles horizontally partitioned data. The goal of the FL framework is to achieve both secure aggregation and differential privacy. This paper adopts a gradient-boosted tree similar to Cheng et al.'s in [5].

Unlike Tian et al., Liu et al. in [7] propose a hybrid FL framework for both VFL and HFL for distributed power

prediction. Power consumption data and weather information are horizontally divided based on district. Demographic data are used for vertical partitioning. The authors claimed to merge horizontal FederBoost and SecureBoost to achieve their goal of a hybrid framework.

Another exciting work that adopts FL is FedDetect [8] by Wen et al. The authors implemented an FL framework based on a Temporal Convolution Network for detecting energy theft in smart grids to preserve data security and privacy. Additionally, a security protocol was established for securing training parameters through homomorphic encryption.

From the few papers mentioned, it can be deduced that most FL frameworks are based on deep learning models. Others that adopt a tree-based model use XGBoost, an implementation of gradient-boosted decision trees. Secondly, most existing tree-based federated learning frameworks work in such a way that all the participating members have to work in tandem to train the global model. The proposed framework is a forest-based FL with disjoint training of the global model. The global model takes advantage of the existing models in the members' infrastructures.

III. PROPOSED FL FRAMEWORK

A. Problem Statement

This study is based on a three-tier hierarchical CIDS architecture proposed in [9]. Each participating member of the collaboration has a specialized IDS trained with a unique dataset. The first two tiers are within the infrastructure of the members, whereas collaboration happens on the topmost tier. The goal is to create an FL framework that will take advantage of the existing IDSs in the local infrastructures while preserving the security and privacy of member data.

B. Concepts

This section explains significant concepts adopted in the security framework.

1) *Assumptions*: There are two assumptions on which the framework relies.

- All the participants of the collaboration adopt a tree-based machine learning algorithm. It can be either a Decision Tree, an Extra Tree, or a forest of either of the trees.
- All the models are properly calibrated, and there is no over-fitting on the local dataset. Capitalizing on existing models requires the capacity to trust the model. The adopted hierarchical architecture relies on prediction probability to trigger collaboration. Therefore, a well-calibrated model improves the ability of the aggregator to trust the model.

2) *Adopted Concepts*: The CIDS architecture adopts a security framework, described in previous work, for secure transmission within the architecture [10]. All transmissions are secured using a hybrid cryptographic algorithm, ensuring that information is secure from third parties, malicious or not.

C. Proposed framework

The proposed framework presents a forest-based FL protocol that takes advantage of the existing classifiers in the member's infrastructure. Assuming there are $M \in \mathbb{N}$ members participating in the collaboration, each has its specialized classifiers $\hat{\Theta}$. Furthermore, each classifier has a range of estimators depending on the type of classifier. To construct the global aggregated model, assume all members adopt a random forest classifier with $K \in \mathbb{N}$ estimators. That is to say that

$$C = \{\hat{\Theta}_1, \dots, \hat{\Theta}_M\}$$

where C is the set of all collaborating classifiers in the architecture, and

$$\hat{\Theta} = \{\hat{\theta}_1, \dots, \hat{\theta}_K\}$$

where $\hat{\theta}$ are estimators in $\hat{\Theta}$.

These estimators are extracted and used to build a new global aggregated forest. It eliminates the complexity of the traditional training of the global model by all the participating members but also introduces the problem of optimizing the forest. Selecting the best estimators for the forest to avoid bias can be challenging. This work focuses on two strategies for selecting the best estimators based on performance. The first strategy adopted is to select the top-performing estimators from the union of all the estimators in the architecture. Members of the collaboration transmit their estimators to the global infrastructure. Each of the estimators is evaluated, and the best n estimators are selected and used to create a new forest.

$$n \leq M \times K, n \in \mathbb{N}$$

The second strategy is the slight equilibrium distribution of estimators among all the participating members of the collaboration (i.e., the number of selected estimators from each of the collaborators is almost equally distributed). A specified number of top estimators $p \in \mathbb{N}$ is selected from each member such that

$$p \times M = n + \mu$$

where μ is a small natural number. μ is added to n to eliminate an equal distribution of estimators among participants. Equality of distribution will negate the voting power of some better-performing estimators in the collaboration. By adding a tunable parameter μ , the influence of each infrastructure is controlled, resolving the favoritism problem in the first strategy. Selecting best p estimators is done using a function f . The function evaluates a set of estimators and returns a given number of best estimators. Each member executes the function to create $\hat{\Theta}_p$, which contains p estimators.

$$\hat{\Theta}_p = f(\hat{\Theta}, p)$$

Next, $\hat{\Theta}_p$ is transmitted to the global layer to form C_* .

$$C_* = \{\hat{\Theta}_{p1}, \dots, \hat{\Theta}_{pM}\}$$

The global layer then executes Algorithm 1 to extract the top n estimators.

Algorithm 1 *Estimators Selection*

Input: C_*, n

Output: $\hat{\Theta}_*$

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1:  $\hat{\Theta}_* := \text{None}$ 
2:  $\hat{\Theta}_c := \text{None}$ 
3: while  $\hat{\Theta}_p \in C_*$  do
4:    $\hat{\Theta}_c := \hat{\Theta}_c \cup \hat{\Theta}_p$ 
5: end while
6:  $\hat{\Theta}_* := f(\hat{\Theta}_c, n)$ 
7: return  $\hat{\Theta}_*$ 

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As shown in Algorithm 1, a union of the estimators is created just like in the first strategy. However, the number of all the collated estimators should be slightly greater than n , which is the expected number of estimators in the new global aggregated forest. A similar function f adopted by the members is also used to extract top n estimators from the collated estimators. f uses a classification challenge to evaluate the estimators to identify the best ones. The performance of an estimator on a challenge is used to grade the estimator, and then further computation is done to select the best-performing estimators.

IV. EVALUATION

Implementation Setup

The test framework was implemented using a Random Forest Classifier as the base model for each participating member. Three candidates were chosen for the testbed, and each of their classifiers has a hundred estimators and was trained on a different dataset. The first participant uses a combination of benign packet data from CAIDA Internet Traces 2008 [11] and attack packet data from CAIDA DDoS Traces 2007 [12]. The second member uses benign packet data, and botnet/DDoS traces from the CICIDS2017 [13] whereas the final participant adopts benign and DoS attack traces from CIC-DoS2017 [14].

Experiment Methodology

As stated in the previous section, the classifier is trained, tested, and calibrated using the local dataset for each participating infrastructure. Afterward, it is tested against a dataset containing samples from all the members of the collaboration. Secondly, a global classifier is trained, tested, and calibrated using a central dataset from all the participating members. Finally, the federated classifier is generated by collating the best-performing trees or estimators from all the participants to create a new forest.

Experiment Results

In order to test the performance of the proposed system, an HFL-based testbed for federated collaboration was implemented, in accordance with Section III-C. First, a calibration evaluation of all the classifiers was conducted to determine their trustworthiness. A calibration plot is used to evaluate the models' calibration. The plot shows a potential mismatch between the model-predicted probabilities (mean predicted values) and the observed probabilities (fraction of positives) in the data. A perfectly calibrated model is illustrated as a straight, diagonal line, representing an identity function (i.e., the predicted and observed probabilities are the same). Hence, the closer the curve is to the line, the better the calibration. Since the framework assumes the classifiers are calibrated, Figure 1 shows the calibration plot.

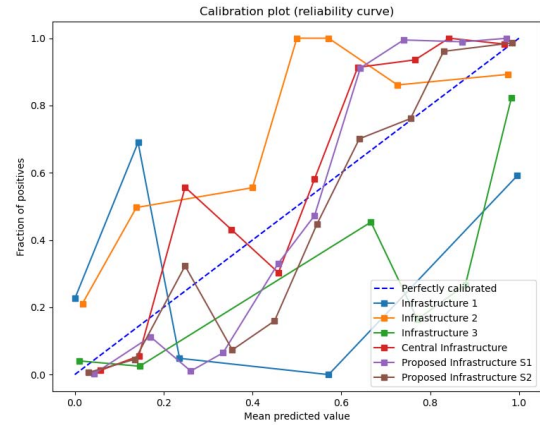


Fig. 1: Model Calibration Test

The plot shows that none of the classifiers are perfectly calibrated; however, some are better calibrated than others. The zigzag shape of most of the lines through the perfect calibration line indicates a fluctuation from overestimation to underestimation. As can be deduced from the curves for the centralized and proposed infrastructures (S1, S2), they started off overestimating probabilities. Next, the centralized and S2 switched to underestimation. Finally, S2 remained the closest to the perfect calibration line compared to the others. The plot also illustrates the ability of the proposed framework to improve model reliability. As seen from the plot, though the central infrastructure was more reliable than the collaborating member, it was not as effective as the proposed framework. This portrays the high reliability of the model generated using the proposed framework. Secondly, the classification performance of the classifiers is evaluated to show the prediction capabilities of the models as depicted in Table I.

Infrastructures 1, 2, and 3 represent members participating in the collaboration, and each has a different dataset. The accuracy shows the performance of the local classifiers on the global dataset, which contains samples unknown to the infrastructure. From the recorded result, it is clear that infrastructure

Infrastructure	Results			
	Accuracy	Precision	Recall	F1-Score
<i>Infrastructure 1</i>	0.64	0.65	0.64	0.63
<i>Infrastructure 2</i>	0.68	0.73	0.69	0.67
<i>Infrastructure 3</i>	0.85	0.88	0.86	0.85
<i>Central Infrastructure</i>	0.89	0.89	0.89	0.89
<i>Proposed Infrastructure S1</i>	0.88	0.89	0.89	0.89
<i>Proposed Infrastructure S2</i>	0.93	0.93	0.93	0.93

TABLE I: Performance of the Classifiers on the Global Test Data

3 outperformed infrastructures 1 and 2, which translates to its dataset having some samples with structures similar to other infrastructures. However, this is not true for infrastructures 1 and 2, which, to a degree, were only able to classify samples from their local infrastructure. The *Central Infrastructure* uses a central dataset (i.e., a combination of the three datasets). As can be deduced from the result, this improves the performance accuracy but, as previously explained, exposes the collaboration to multiple security concerns. Finally, the strategies for selecting the best estimators were evaluated. As can be seen from the table, the *Proposed Infrastructure S1* performed just as well as the *Central Infrastructure* with a prediction accuracy of 88%. All estimators were collected in this infrastructure, and the best n estimators were chosen. Some collaborators may be concerned about security because the entire structure of their network is being transmitted. Nonetheless, this is resolved using the second strategy. *Proposed Infrastructure S2* outperformed all the other classifiers with a prediction accuracy of 93%. This is because it allows for better participation from all the participants in the collaboration. On the contrary, the first strategy allows for unequal participation among the participants because an infrastructure with better-performing estimators will have a higher representation in the forest, thereby reducing the capability of the final classifier to predict samples from the less represented infrastructure. Therefore, a slight equilibrium distribution of the estimators is more advantageous than favoring higher-performing infrastructures.

V. CONCLUSION

The proposed framework is a CIDS privacy-preserving architecture with a forest-based Federated Learning scheme. A new FL framework is proposed to eliminate the need for transmitting sensitive network information in the architecture to facilitate collaboration. The framework requires the construction of a new aggregated forest from existing tree-based classifiers without the need for tandem training by all the participating members of the collaboration. To obtain this, two strategies were proposed. Firstly, the union of the estimators from existing tree-based classifiers belonging to participating members of the collaboration and then selecting the best-performing estimators from the union. This encourages favoritism in selecting the best estimators because infrastructures with better-performing estimators contribute more to the forest than others. It also increases the chances that some estimators will not be represented in the forest. The second strategy is a slight equilibrium, or almost equal distribution, of estimators

in the forest. It means that all the participants send an equal number of their best-performing estimators to the aggregator, which then selects the needed estimators from the reduced set. It encourages the participation of all members in the new forest and outperforms the previous strategy. Based on the result of the experiment, it is clear that the proposed FL strategies outperformed the centralized model.

This paper presents the possibility of a forest-based Federated Learning framework. Nevertheless, it is necessary to establish the need for further investigation into optimizing the forest. Although the two strategies presented were effective, they left room for further improvement.

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